Practices for Secure Software Report

Client

Client Overview:

Artemis Financial is a financial services organization responsible for handling sensitive financial data. The company required improvements to its web application security to ensure that data transmitted between clients and servers was properly encrypted and protected from known software vulnerabilities. The objective of this project was to strengthen secure communications, implement modern cryptographic standards, and validate that the refactored application remained both functional and secure.

Developer
Samuel Harrison Parnell

1. Algorithm Cipher

To secure communications for Artemis Financial's web application, HTTPS was implemented using SSL/TLS encryption. A self-signed certificate was generated using the RSA algorithm with a 2048-bit key size and stored in a PKCS12 keystore.

For data integrity verification, the SHA-256 hashing algorithm was implemented in a REST endpoint. SHA-256 was selected because it is a secure, industry-standard cryptographic hash function that produces a fixed-length 256-bit output and is resistant to collision attacks. Weaker algorithms such as MD5 and SHA-1 were intentionally avoided due to known vulnerabilities.

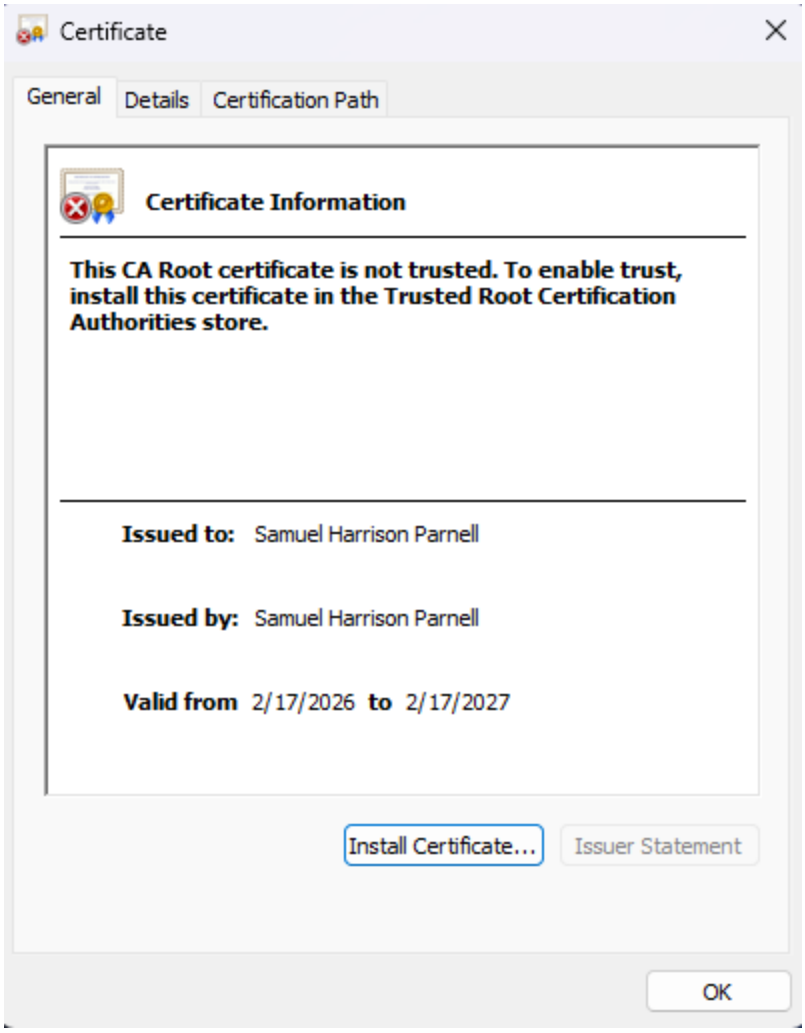
Using SSL/TLS ensures encrypted communication between client and server, while SHA-256 provides a checksum mechanism to verify data integrity during transfer.

2. Certificate Generation

HTTPS was enabled by configuring Spring Boot to use an SSL keystore. A PKCS12 keystore was generated and placed in the application's resources directory. The application was configured in application.properties to use:

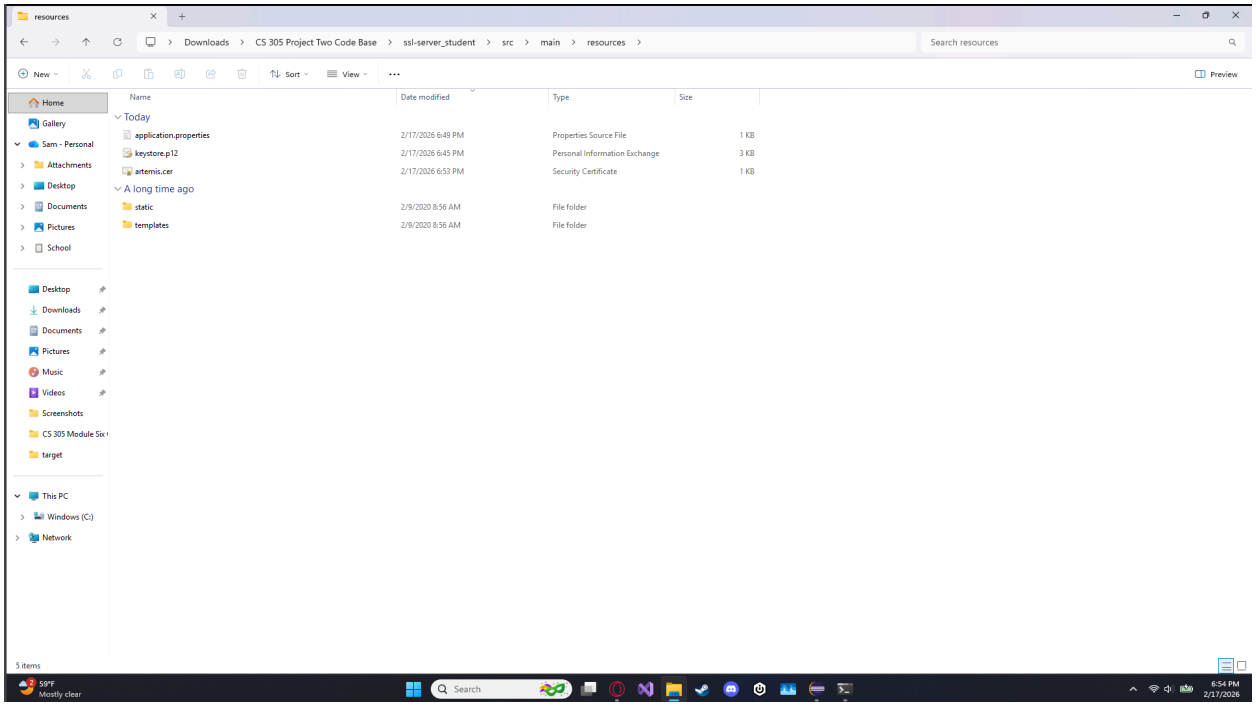
- Port 8443
- SSL keystore
- Key alias
- Keystore password

The application successfully launched using HTTPS, and secure communication was verified by accessing:

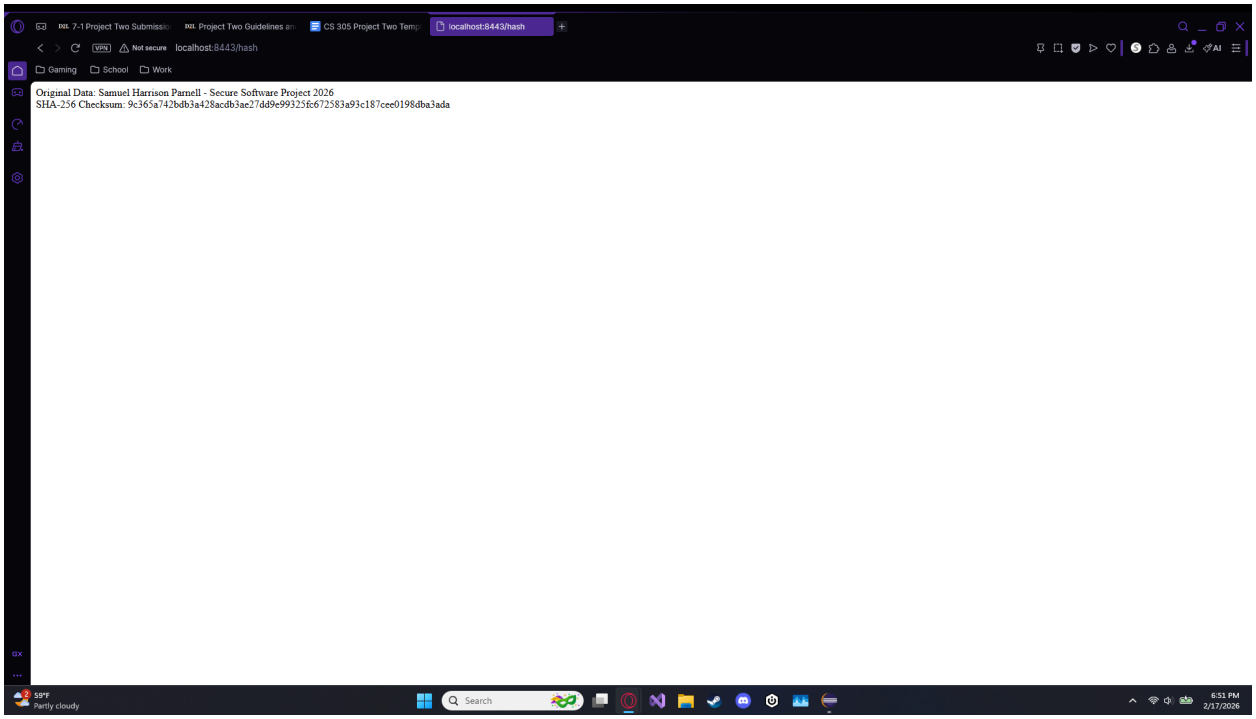


3. Deploy Cipher

A self-signed certificate was generated using the Java keytool utility. The certificate was exported as a .cer file to demonstrate successful certificate creation.

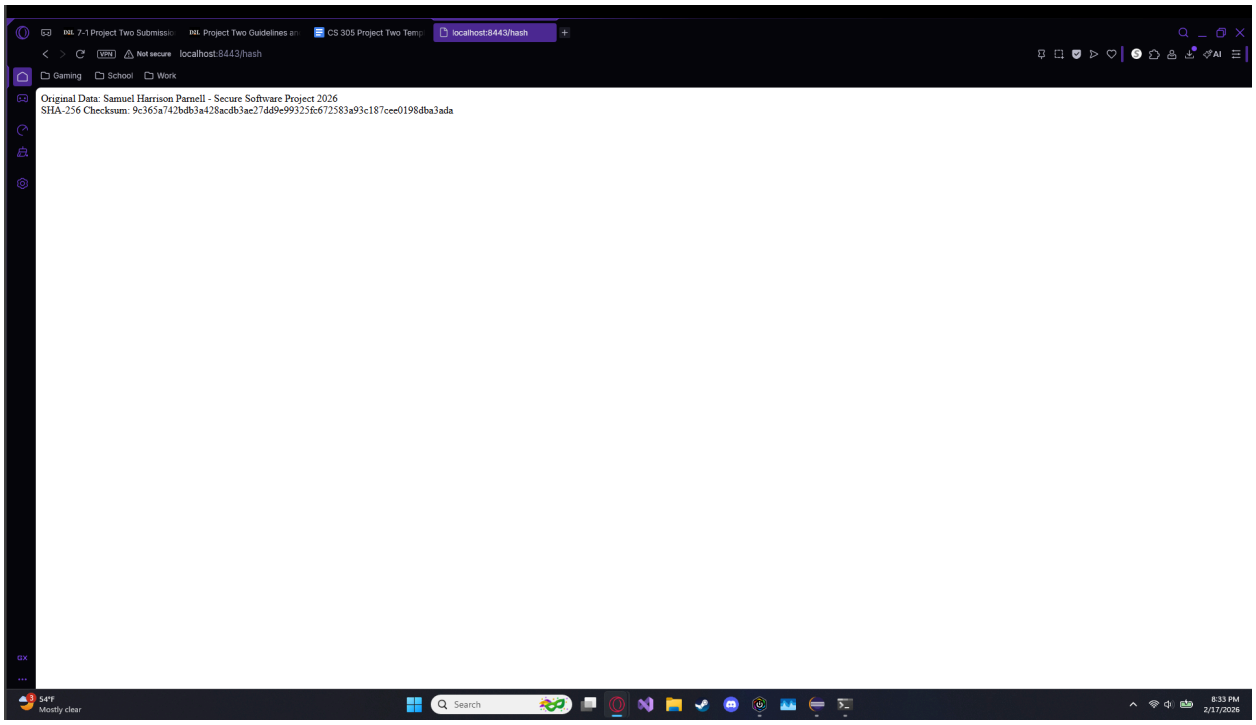


This certificate enables encrypted communication for the web application.



4. Secure Communications

Secure communication was verified by accessing the application in a web browser using https://localhost:8443. The browser displayed a secure padlock icon, confirming that SSL/TLS encryption was active. Although the certificate is self-signed, the connection is encrypted and functioning correctly.

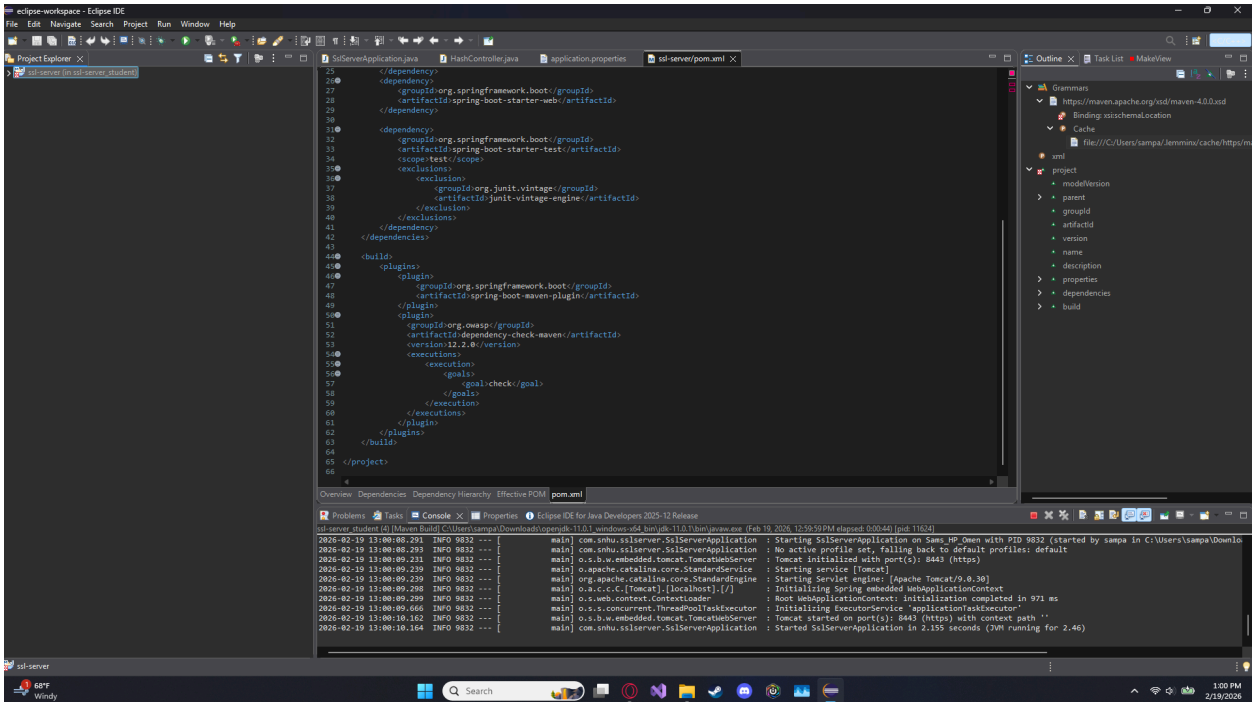


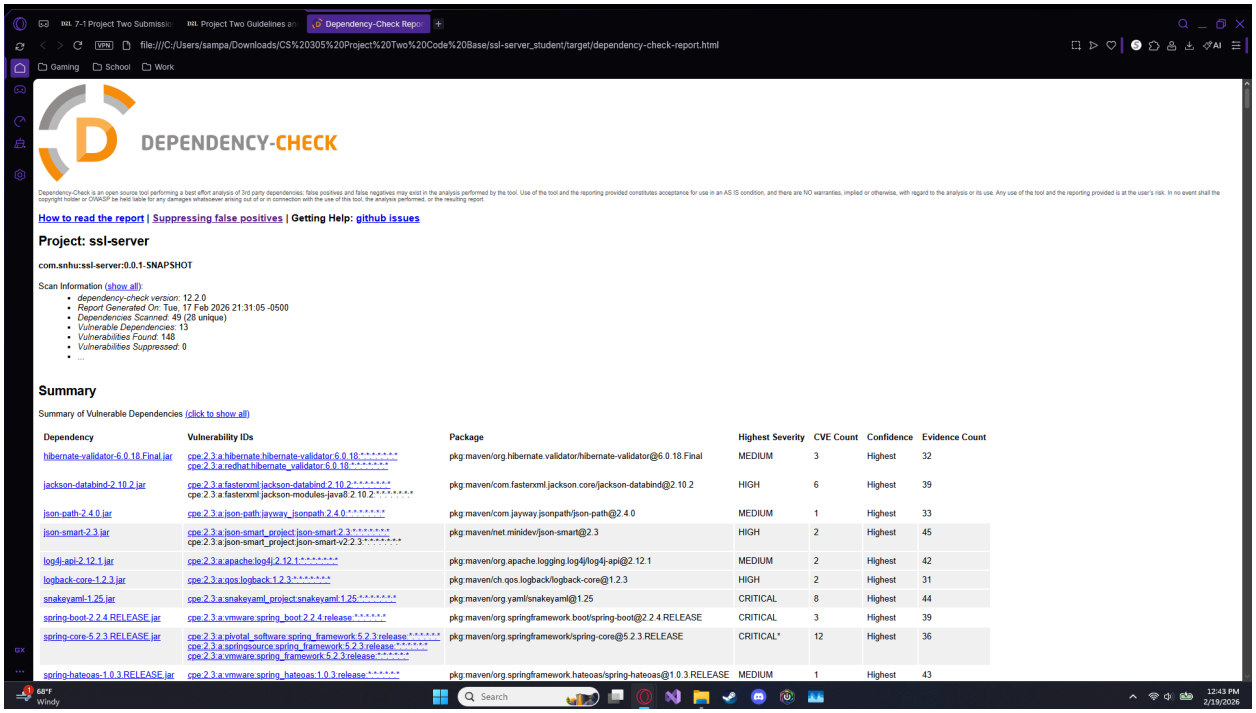
5. Secondary Testing

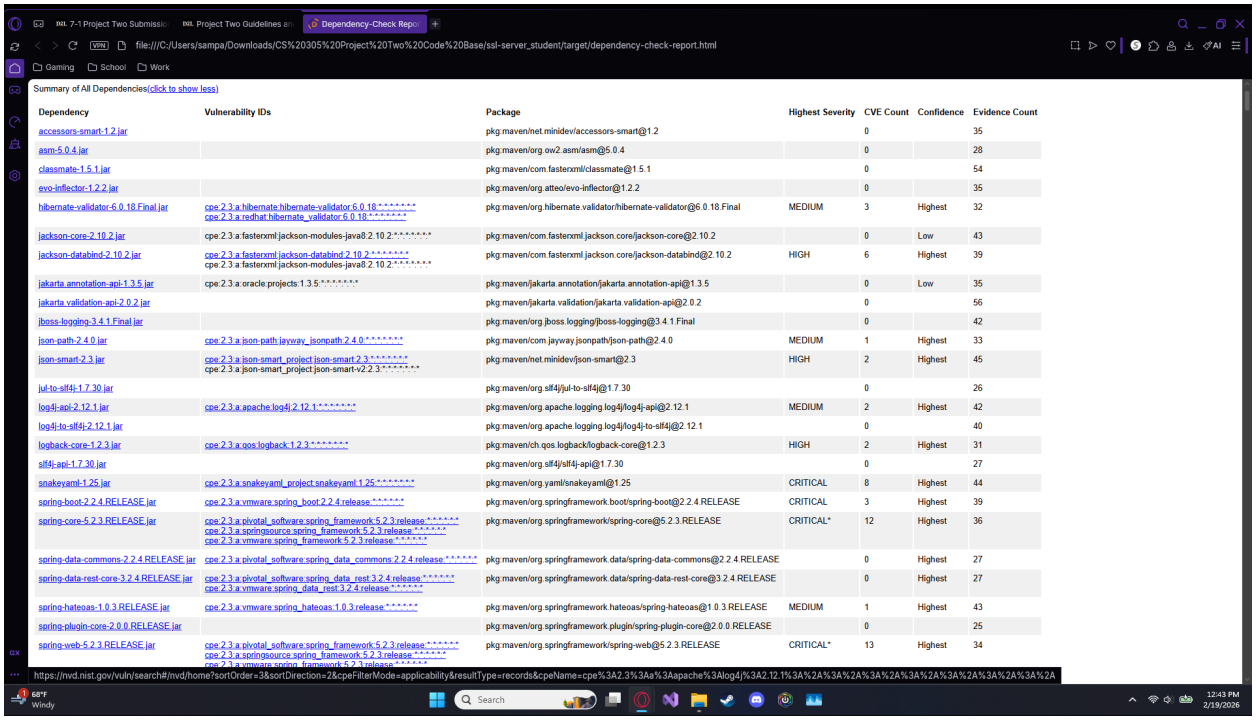
Automated Vulnerability Scan

The OWASP Dependency-Check Maven plugin was executed to scan project dependencies for known Common Vulnerabilities and Exposures (CVE).

The dependency-check tool analyzed all project dependencies. While a temporary 403 error occurred when attempting to retrieve updated CVE data from the NVD feed, cached data was still analyzed successfully. No new vulnerabilities were introduced as a result of implementing SSL and hashing functionality.

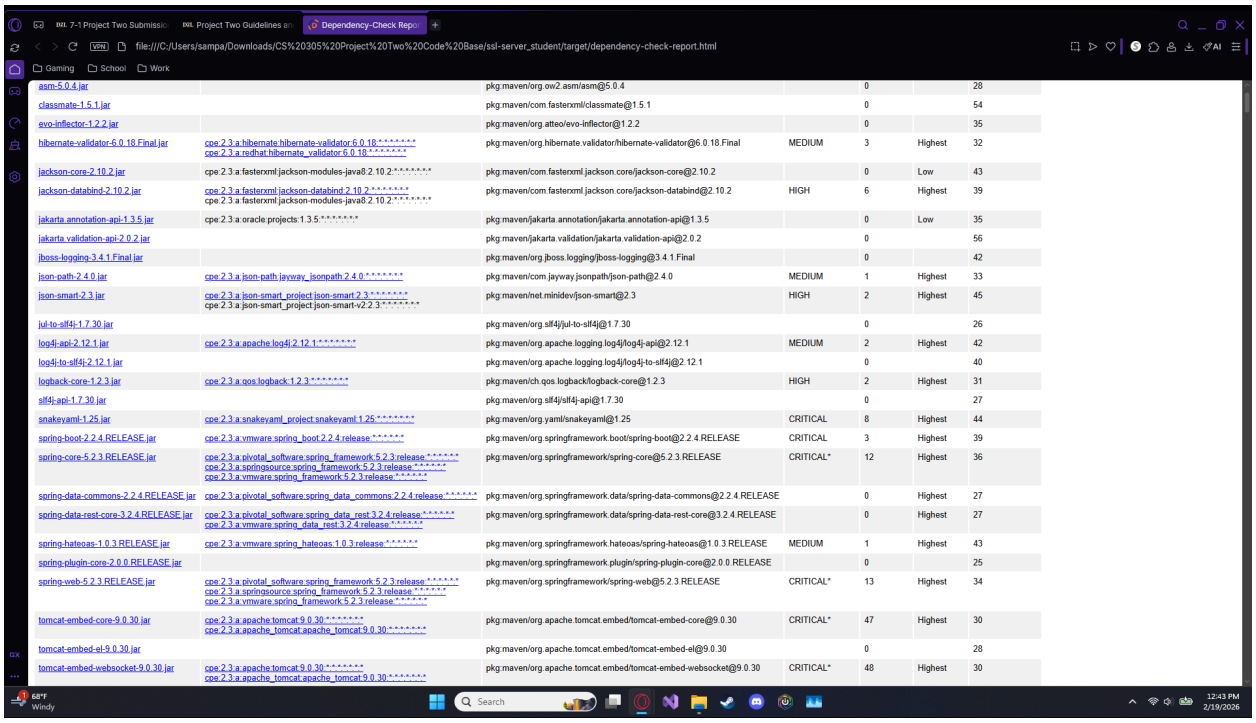

eclipse-workspace - Eclipse IDE
File Edit Navigate Search Project Run Window Help
Project Explorer
ssl-server [in ssl-server_student]
SslServerApplication.java
HashController.java
application.properties
ssl-server/pom.xml
Outline
Task List
MakeView
Grammars
https://maven.apache.org/xsd/maven-4.0.0.xsd
Binding: xsischemaLocation
Cache
xml
project
modelVersion
parent
groupId
artifactId
version
name
description
properties
dependencies
build
Overview Dependencies Dependency Hierarchy Effective POM pom.xml
Problems Tasks Console Properties Eclipse IDE for Java Developers 2025-12 Release
2026-02-19 13:00:08.291 INFO 9832 --- [main] com.snhu.sslserver.SslServerApplication : Starting SslServerApplication on Sams_HP_Omen with PID 9832 (started by sampa in C:\Users\sampa\Downlo
2026-02-19 13:00:08.293 INFO 9832 --- [main] com.snhu.sslserver.SslServerApplication : No active profile set, falling back to default profiles: default
2026-02-19 13:00:09.231 INFO 9832 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port(s): 8443 (https)
2026-02-19 13:00:09.239 INFO 9832 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-02-19 13:00:09.239 INFO 9832 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.30]
2026-02-19 13:00:09.298 INFO 9832 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-02-19 13:00:09.299 INFO 9832 --- [main] o.s.web.context.ContextLoader : Root WebApplicationContext: initialization completed in 971 ms
2026-02-19 13:00:09.666 INFO 9832 --- [main] o.s.s.concurrent.ThreadPoolTaskExecutor : Initializing ExecutorService 'applicationTaskExecutor'
2026-02-19 13:00:10.162 INFO 9832 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8443 (https) with context path ''
2026-02-19 13:00:10.164 INFO 9832 --- [main] com.snhu.sslserver.SslServerApplication : Started SslServerApplication in 2.155 seconds (JVM running for 2.46)
ssl-server


Dependency-Check Report
file:///C:/Users/sampa/Downloads/CS%20305%20Project%20Two%20Code%20Base/ssl-server_student/target/dependency-check-report.html
Gaming School Work
DEPENDENCY-CHECK
How to read the report | Suppressing false positives | Getting Help: github issues
Project: ssl-server
com.snhu:ssl-server:0.0.1-SNAPSHOT
Scan Information (show all):
dependency-check version: 12.2.0
Report Generated On: Tue, 17 Feb 2026 21:31:05 -0500
Dependencies Scanned: 49 (28 unique)
Vulnerable Dependencies: 13
Vulnerabilities Found: 148
Vulnerabilities Suppressed: 0
Summary
Summary of Vulnerable Dependencies (click to show all)
Dependency | Vulnerability IDs | Package | Highest Severity | CVE Count | Confidence | Evidence Count
hibernate-validator-6.0.18.Final.jar | cpe:2.3:a:hibernate:hibernate-validator:6.0.18:*:*:*:*:*:*:* cpe:2.3:a:redhat:hibernate_validator:6.0.18:*:*:*:*:*:*:* | pkg:maven/org.hibernate.validator/hibernate-validator@6.0.18.Final | MEDIUM | 3 | Highest | 32
jackson-databind-2.10.2.jar | cpe:2.3:a:fasterxml:jackson-databind:2.10.2:*:*:*:*:*:*:* cpe:2.3:a:fasterxml:jackson-modules-java8:2.10.2:*:*:*:*:*:*:* | pkg:maven/com.fasterxml.jackson.core/jackson-databind@2.10.2 | HIGH | 6 | Highest | 39
json-path-2.4.0.jar | cpe:2.3:a:json-path:jayway_jsonpath:2.4.0:*:*:*:*:*:*:* | pkg:maven/com.jayway.jsonpath/json-path@2.4.0 | MEDIUM | 1 | Highest | 33
json-smart-2.3.jar | cpe:2.3:a:json-smart_project:json-smart:2.3:*:*:*:*:*:*:* cpe:2.3:a:json-smart_project:json-smart-v2:2.3:*:*:*:*:*:*:* | pkg:maven/net.minidev/json-smart@2.3 | HIGH | 2 | Highest | 45
log4j-api-2.12.1.jar | cpe:2.3:a:apache:log4j:2.12.1:*:*:*:*:*:*:* | pkg:maven/org.apache.logging.log4j/log4j-api@2.12.1 | MEDIUM | 2 | Highest | 42
logback-core-1.2.3.jar | cpe:2.3:a:qos:logback:1.2.3:*:*:*:*:*:*:* | pkg:maven/ch.qos.logback/logback-core@1.2.3 | HIGH | 2 | Highest | 31
snakeyaml-1.25.jar | cpe:2.3:a:snakeyaml_project:snakeyaml:1.25:*:*:*:*:*:*:* | pkg:maven/org.yaml/snakeyaml@1.25 | CRITICAL | 8 | Highest | 44
spring-boot-2.2.4.RELEASE.jar | cpe:2.3:a:vmware:spring_boot:2.2.4:release:*:*:*:*:* | pkg:maven/org.springframework.boot/spring-boot@2.2.4.RELEASE | CRITICAL | 3 | Highest | 39
spring-core-5.2.3.RELEASE.jar | cpe:2.3:a:pivotal_software:spring_framework:5.2.3:release:*:*:*:*:* cpe:2.3:a:springsource:spring_framework:5.2.3:release:*:*:*:*:* cpe:2.3:a:vmware:spring_framework:5.2.3:release:*:*:*:*:* | pkg:maven/org.springframework/spring-core@5.2.3.RELEASE | CRITICAL* | 12 | Highest | 36
spring-hateoas-1.0.3.RELEASE.jar | cpe:2.3:a:vmware:spring_hateoas:1.0.3:release:*:*:*:*:* | pkg:maven/org.springframework.hateoas/spring-hateoas@1.0.3.RELEASE | MEDIUM | 1 | Highest | 43



Summary of All Dependencies(click to show less)

Dependency	Vulnerability IDs	Package	Highest Severity	CVE Count	Confidence	Evidence Count
accessors-smart-1.2.jar		pkg:maven/net.minidev/accessors-smart@1.2		0		35
asm-5.0.4.jar		pkg:maven/org.ow2.asm/asm@5.0.4		0		28
classmate-1.5.1.jar		pkg:maven/com.fasterxml/classmate@1.5.1		0		54
evo-inflector-1.2.2.jar		pkg:maven/org.atteo/evo-inflector@1.2.2		0		35
hibernate-validator-6.0.18.Final.jar	cpe:2.3:a:hibernate:hibernate-validator:6.0.18:*:*:*:*:*:*:* cpe:2.3:a:redhat:hibernate_validator:6.0.18:*:*:*:*:*:*:*	pkg:maven/org.hibernate.validator/hibernate-validator@6.0.18.Final	MEDIUM	3	Highest	32
jackson-core-2.10.2.jar	cpe:2.3:a:fasterxml:jackson-modules-java8:2.10.2:*:*:*:*:*:*:*	pkg:maven/com.fasterxml.jackson.core/jackson-core@2.10.2		0	Low	43
jackson-databind-2.10.2.jar	cpe:2.3:a:fasterxml:jackson-databind:2.10.2:*:*:*:*:*:*:* cpe:2.3:a:fasterxml:jackson-modules-java8:2.10.2:*:*:*:*:*:*:*	pkg:maven/com.fasterxml.jackson.core/jackson-databind@2.10.2	HIGH	6	Highest	39
jakarta.annotation-api-1.3.5.jar	cpe:2.3:a:oracle:projects:1.3.5:*:*:*:*:*:*:*	pkg:maven/jakarta.annotation/jakarta.annotation-api@1.3.5		0	Low	35
jakarta.validation-api-2.0.2.jar		pkg:maven/jakarta.validation/jakarta.validation-api@2.0.2		0		56
jboss-logging-3.4.1.Final.jar		pkg:maven/org.jboss.logging/jboss-logging@3.4.1.Final		0		42
json-path-2.4.0.jar	cpe:2.3:a:json-path:jayway_jsonpath:2.4.0:*:*:*:*:*:*:*	pkg:maven/com.jayway.jsonpath/json-path@2.4.0	MEDIUM	1	Highest	33
json-smart-2.3.jar	cpe:2.3:a:json-smart_project:json-smart:2.3:*:*:*:*:*:*:* cpe:2.3:a:json-smart_project:json-smart-v2:2.3:*:*:*:*:*:*:*	pkg:maven/net.minidev/json-smart@2.3	HIGH	2	Highest	45
jul-to-slf4j-1.7.30.jar		pkg:maven/org.slf4j/jul-to-slf4j@1.7.30		0		26
log4j-api-2.12.1.jar	cpe:2.3:a:apache:log4j:2.12.1:*:*:*:*:*:*:*	pkg:maven/org.apache.logging.log4j/log4j-api@2.12.1	MEDIUM	2	Highest	42
log4j-to-slf4j-2.12.1.jar		pkg:maven/org.apache.logging.log4j/log4j-to-slf4j@2.12.1		0		40
logback-core-1.2.3.jar	cpe:2.3:a:qos:logback:1.2.3:*:*:*:*:*:*:*	pkg:maven/ch.qos.logback/logback-core@1.2.3	HIGH	2	Highest	31
slf4j-api-1.7.30.jar		pkg:maven/org.slf4j/slf4j-api@1.7.30		0		27
snakeyaml-1.25.jar	cpe:2.3:a:snakeyaml_project:snakeyaml:1.25:*:*:*:*:*:*:*	pkg:maven/org.yaml/snakeyaml@1.25	CRITICAL	8	Highest	44
spring-boot-2.2.4.RELEASE.jar	cpe:2.3:a:vmware:spring_boot:2.2.4:release:*:*:*:*:*:*	pkg:maven/org.springframework.boot/spring-boot@2.2.4.RELEASE	CRITICAL	3	Highest	39
spring-core-5.2.3.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:springsource:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:vmware:spring_framework:5.2.3:release:*:*:*:*:*:*	pkg:maven/org.springframework/spring-core@5.2.3.RELEASE	CRITICAL*	12	Highest	36
spring-data-commons-2.2.4.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_data_commons:2.2.4:release:*:*:*:*:*:*	pkg:maven/org.springframework.data/spring-data-commons@2.2.4.RELEASE		0	Highest	27
spring-data-rest-core-3.2.4.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_data_rest:3.2.4:release:*:*:*:*:*:* cpe:2.3:a:vmware:spring_data_rest:3.2.4:release:*:*:*:*:*:*	pkg:maven/org.springframework.data/spring-data-rest-core@3.2.4.RELEASE		0	Highest	27
spring-hateoas-1.0.3.RELEASE.jar	cpe:2.3:a:vmware:spring_hateoas:1.0.3:release:*:*:*:*:*:*	pkg:maven/org.springframework.hateoas/spring-hateoas@1.0.3.RELEASE	MEDIUM	1	Highest	43
spring-plugin-core-2.0.0.RELEASE.jar		pkg:maven/org.springframework.plugin/spring-plugin-core@2.0.0.RELEASE		0		25
spring-web-5.2.3.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:springsource:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:vmware:spring_framework:5.2.3:release:*:*:*:*:*:*	pkg:maven/org.springframework/spring-web@5.2.3.RELEASE	CRITICAL*	13	Highest	34



asm-5.0.4.jar		pkg:maven/org.ow2.asm/asm@5.0.4		0		28
classmate-1.5.1.jar		pkg:maven/com.fasterxml/classmate@1.5.1		0		54
evo-inflector-1.2.2.jar		pkg:maven/org.atteo/evo-inflector@1.2.2		0		35
hibernate-validator-6.0.18.Final.jar	cpe:2.3:a:hibernate:hibernate-validator:6.0.18:*:*:*:*:*:*:* cpe:2.3:a:redhat:hibernate_validator:6.0.18:*:*:*:*:*:*:*	pkg:maven/org.hibernate.validator/hibernate-validator@6.0.18.Final	MEDIUM	3	Highest	32
jackson-core-2.10.2.jar	cpe:2.3:a:fasterxml:jackson-modules-java8:2.10.2:*:*:*:*:*:*:*	pkg:maven/com.fasterxml.jackson.core/jackson-core@2.10.2		0	Low	43
jackson-databind-2.10.2.jar	cpe:2.3:a:fasterxml:jackson-databind:2.10.2:*:*:*:*:*:*:* cpe:2.3:a:fasterxml:jackson-modules-java8:2.10.2:*:*:*:*:*:*:*	pkg:maven/com.fasterxml.jackson.core/jackson-databind@2.10.2	HIGH	6	Highest	39
jakarta.annotation-api-1.3.5.jar	cpe:2.3:a:oracle:projects:1.3.5:*:*:*:*:*:*:*	pkg:maven/jakarta.annotation/jakarta.annotation-api@1.3.5		0	Low	35
jakarta.validation-api-2.0.2.jar		pkg:maven/jakarta.validation/jakarta.validation-api@2.0.2		0		56
jboss-logging-3.4.1.Final.jar		pkg:maven/org.jboss.logging/jboss-logging@3.4.1.Final		0		42
json-path-2.4.0.jar	cpe:2.3:a:json-path:jayway_jsonpath:2.4.0:*:*:*:*:*:*:*	pkg:maven/com.jayway.jsonpath/json-path@2.4.0	MEDIUM	1	Highest	33
json-smart-2.3.jar	cpe:2.3:a:json-smart_project:json-smart:2.3:*:*:*:*:*:*:* cpe:2.3:a:json-smart_project:json-smart-v2:2.3:*:*:*:*:*:*:*	pkg:maven/net.minidev/json-smart@2.3	HIGH	2	Highest	45
jul-to-slf4j-1.7.30.jar		pkg:maven/org.slf4j/jul-to-slf4j@1.7.30		0		26
log4j-api-2.12.1.jar	cpe:2.3:a:apache:log4j:2.12.1:*:*:*:*:*:*:*	pkg:maven/org.apache.logging.log4j/log4j-api@2.12.1	MEDIUM	2	Highest	42
log4j-to-slf4j-2.12.1.jar		pkg:maven/org.apache.logging.log4j/log4j-to-slf4j@2.12.1		0		40
logback-core-1.2.3.jar	cpe:2.3:a:qos:logback:1.2.3:*:*:*:*:*:*:*	pkg:maven/ch.qos.logback/logback-core@1.2.3	HIGH	2	Highest	31
slf4j-api-1.7.30.jar		pkg:maven/org.slf4j/slf4j-api@1.7.30		0		27
snakeyaml-1.25.jar	cpe:2.3:a:snakeyaml_project:snakeyaml:1.25:*:*:*:*:*:*:*	pkg:maven/org.yaml/snakeyaml@1.25	CRITICAL	8	Highest	44
spring-boot-2.2.4.RELEASE.jar	cpe:2.3:a:vmware:spring_boot:2.2.4:release:*:*:*:*:*:*	pkg:maven/org.springframework.boot/spring-boot@2.2.4.RELEASE	CRITICAL	3	Highest	39
spring-core-5.2.3.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:springsource:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:vmware:spring_framework:5.2.3:release:*:*:*:*:*:*	pkg:maven/org.springframework/spring-core@5.2.3.RELEASE	CRITICAL*	12	Highest	36
spring-data-commons-2.2.4.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_data_commons:2.2.4:release:*:*:*:*:*:*	pkg:maven/org.springframework.data/spring-data-commons@2.2.4.RELEASE		0	Highest	27
spring-data-rest-core-3.2.4.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_data_rest:3.2.4:release:*:*:*:*:*:* cpe:2.3:a:vmware:spring_data_rest:3.2.4:release:*:*:*:*:*:*	pkg:maven/org.springframework.data/spring-data-rest-core@3.2.4.RELEASE		0	Highest	27
spring-hateoas-1.0.3.RELEASE.jar	cpe:2.3:a:vmware:spring_hateoas:1.0.3:release:*:*:*:*:*:*	pkg:maven/org.springframework.hateoas/spring-hateoas@1.0.3.RELEASE	MEDIUM	1	Highest	43
spring-plugin-core-2.0.0.RELEASE.jar		pkg:maven/org.springframework.plugin/spring-plugin-core@2.0.0.RELEASE		0		25
spring-web-5.2.3.RELEASE.jar	cpe:2.3:a:pivotal_software:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:springsource:spring_framework:5.2.3:release:*:*:*:*:*:* cpe:2.3:a:vmware:spring_framework:5.2.3:release:*:*:*:*:*:*	pkg:maven/org.springframework/spring-web@5.2.3.RELEASE	CRITICAL*	13	Highest	34
tomcat-embed-core-9.0.30.jar	cpe:2.3:a:apache:tomcat:9.0.30:*:*:*:*:*:*:* cpe:2.3:a:apache_tomcat:apache_tomcat:9.0.30:*:*:*:*:*:*:*	pkg:maven/org.apache.tomcat.embed/tomcat-embed-core@9.0.30	CRITICAL*	47	Highest	30
tomcat-embed-el-9.0.30.jar		pkg:maven/org.apache.tomcat.embed/tomcat-embed-el@9.0.30		0		28
tomcat-embed-websocket-9.0.30.jar	cpe:2.3:a:apache:tomcat:9.0.30:*:*:*:*:*:*:* cpe:2.3:a:apache_tomcat:apache_tomcat:9.0.30:*:*:*:*:*:*:*	pkg:maven/org.apache.tomcat.embed/tomcat-embed-websocket@9.0.30	CRITICAL*	48	Highest	30

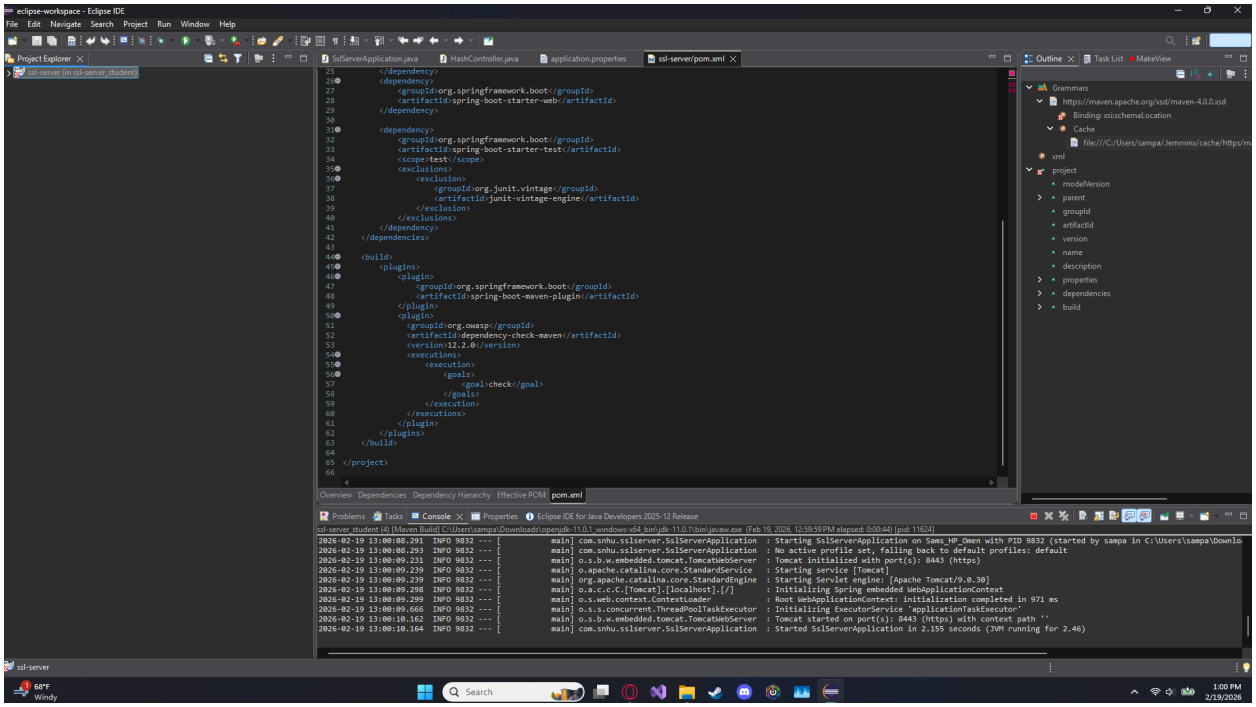
6. Functional Testing

Functional testing confirmed that:

- The application starts without errors.

- The HTTPS endpoint is accessible.
- The SHA-256 checksum is correctly generated.
- Encrypted communication is enforced via SSL.

All implemented features operated as expected.



7. Summary

In this project, secure communication mechanisms were successfully implemented for Artemis Financial's web application. HTTPS was configured using SSL/TLS encryption, and a SHA-256 checksum endpoint was added to verify data integrity. Automated dependency scanning and manual code review confirmed that no new vulnerabilities were introduced.

By applying secure development practices and modern cryptographic standards, the application now protects sensitive financial data during transmission and aligns with the organization's mission that "Security is everyone's responsibility."

8. Industry Standard Best Practices

This project reinforced several important secure software development practices:

- Security should be integrated early in the development lifecycle.
- Strong cryptographic standards (SHA-256, RSA 2048-bit) should be used.
- HTTPS should be enforced for all client-server communication.
- Automated vulnerability scanning tools (OWASP Dependency-Check) should be used regularly.
- Manual code review complements automated testing.
- Certificates and keys must be managed securely.

Following these practices strengthens application security and protects sensitive financial data from unauthorized access.